

PICUFLA

Functional and Non-functional Requirements

Section 1: Functional Requirements

Onboarding (First Time Use)

- The app shall show a welcome screen that introduces the app when a user opens it for the first time.
- The app shall show a screen that lists the main features of the app during first-time setup.
- The app shall ask the user for permission to use the camera before the user can take a photo.
- The app shall ask the user for permission to access their photo library before the user can pick a photo.
- The app shall show the privacy policy to the user during onboarding.
- The app shall require the user to accept the privacy policy before they can continue using the app.

Account Management

- The app shall let users create an account using their email address and a password.
- The app shall let users log in using their email address and password.
- The app shall let users log out of their account at any time.
- The app shall offer a "forgot password" option that sends a code to the user's email so they can reset their password.
- The app shall let users change their password while logged in.
- The app shall let users set up a profile with a display name, a short bio, and a profile photo.
- The app shall let users edit their profile information at any time.
- The app shall let users delete their account, which removes all of their personal data.

Security

- The app shall ask the user to re-verify their email every 12 hours using a code sent to their inbox.
- The app shall slow down repeated login attempts if a user enters the wrong password too many times, starting with a 30-second wait and increasing up to 5 minutes.

Plant Identification

- The app shall let users take a photo of a plant using their phone's camera.
- The app shall let users pick an existing photo of a plant from their phone's gallery.
- The app shall let users turn the camera flash on or off while taking a photo.
- The app shall send the photo to the plant identification system so the plant can be recognized.

- The app shall show the identification result, including the plant's common name, scientific name, how confident the system is in the result, and care information covering watering, sunlight, soil, and uses.
- The app shall show alternative possible plant matches if the system is less than 60% confident in its first answer.
- The app shall limit plant identification to 5 uses per hour for each user.

Plant Collection

- The app shall let users save an identified plant to their personal collection.
- The app shall show all of a user's saved plants in a grid layout.
- The app shall let users search their collection by plant name.
- The app shall let users sort their collection by newest, oldest, A to Z, or Z to A.
- The app shall let users filter their collection by tags they have added.
- The app shall let users mark any plant in their collection as a favorite.
- The app shall let users remove a plant from their collection, with the option to recover it later.
- The app shall update the collection right away whenever a plant is added, changed, or removed.

Plant Details

- The app shall show full plant information, including common name, scientific name, description, and care guide.
- The app shall show care guide sections covering watering, sunlight, soil, uses, and helpful tips.
- The app shall let users add and edit personal notes for a plant, up to 500 characters.
- The app shall let users add and remove tags for a plant, up to 10 tags per plant.
- The app shall let users manage reminders for that specific plant from the plant's detail screen.

Reminders

- The app shall let users create a reminder for a plant, choosing from water, fertilize, repot, or a custom reminder.
- The app shall let users pick a date and time for each reminder.
- The app shall let users set a reminder to repeat daily, weekly, or monthly.
- The app shall show all reminders that have been set for a plant.
- The app shall let users delete a reminder.
- The app shall send a notification to the user's phone when a reminder is due.
- The app shall ask the user for permission before sending any notifications.

Profile & Settings

- The app shall show the user's profile, including their avatar, display name, and email.
- The app shall show the user's stats, including how many plants they have discovered, how many favorites they have, and how many reminders they have set.
- The app shall let users turn notifications on or off.

- The app shall let users enable or disable offline mode, which allows them to view their collection without an internet connection.
- The app shall let users view the privacy policy at any time from settings.
- The app shall let users log out from the settings screen.

Offline Support

- The app shall let users view their saved plant collection even when offline mode is turned on and there is no internet connection.
- The app shall automatically sync any changes once the internet connection is restored.

Data Privacy

- The app shall let users request the deletion of all their data at any time.
 - The app shall permanently remove deleted data within 30 days of the request.
 - The app shall keep a record of what data was deleted and when, for accountability purposes.
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Section 2: Non-functional Requirements

Usability

- The app shall feel simple and easy to use, even for someone using it for the first time.
- The app shall load screens quickly, without making the user wait a long time.
- The app shall show error messages in plain, everyday language instead of technical codes.
- The app shall guide the user step by step through key processes, such as onboarding.

Performance

- The app shall respond to taps and swipes within 1 second.
- The app shall complete a plant identification, from photo to result, within 15 seconds.
- The app shall run smoothly without crashing or freezing during normal use.

Reliability

- The app shall handle a lost internet connection gracefully by showing a clear message instead of crashing.
- The app shall never lose a user's saved data, including their collection and reminders.
- The app shall work correctly on both iPhone and Android devices.

Security & Privacy

- The app shall never show or store a user's password in plain, readable text.
- The app shall require the user to periodically re-verify their identity to keep their session secure.
- The app shall keep the key used to access the plant identification system stored safely on the server, never on the user's phone.

- The app shall only allow a user to view or change their own data, never another user's data.
- The app shall keep each user's photos private, visible only to that user.

Availability

- The app shall be available for use at any time, day or night.
- The app shall inform users in advance of any planned maintenance, whenever possible.

Compatibility

- The app shall work on iPhone models running iOS 16 or newer, and Android phones running Android 13 or newer.
- The app shall work properly on both phones and tablets.